

Tutorial 6: AI-Assisted Risk Modelling and Diversification in R

Student Exploration: Simulation, Correlation and Diversification

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Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

1 Dependence: Covariance, Correlation and the Hidden Link Between Risks

Mini-Mission: Can two risky things become safer together?

You will explore how two uncertain quantities move together, and why this matters for diversification.

Key Definition / Result

For two random variables X and Y ,

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

The correlation coefficient is

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho_{X,Y} \leq 1.$$

Covariance and correlation tell us whether two uncertain quantities tend to move together, move in opposite directions, or have little linear co-movement.

Suggested AI Prompt: Simulating Correlated Returns *** Advanced

"Act as a quantitative finance tutor. Please write beginner-friendly R code to simulate two asset returns with a chosen correlation (e.g., -0.5). Since I only know the Uniform distribution, use runif() to generate the base returns and independent noise. Then calculate the sample correlation and plot the points."

Predict Before Coding: What will a negative correlation look like?

Before plotting, sketch the scatter plot for $\rho = -0.5$. Should the cloud slope upward, downward, or show no clear pattern?

Investigation / Activity: Creating Two Artificial Assets With Chosen Correlation

*** Advanced

Idea: Instead of relying on real market data, we create artificial return data using the Uniform distribution. By carefully mixing a shared variable and an independent variable, we can control the correlation!

R Code:

```
set.seed(2026)
number_of_days <- 1000
chosen_correlation <- -0.5

# Step 1: Create a base sequence of uniform returns for Asset A.
# We center it at 0 so our correlation mixing math works cleanly.
asset_A_centered <- runif(
  n = number_of_days,
  min = -0.02,
  max = 0.02
)

# Step 2: Create independent noise using the same uniform spread.
independent_noise <- runif(
  n = number_of_days,
  min = -0.02,
  max = 0.02
)

# Step 3: Create Asset B by mixing Asset A and the noise.
# This mathematical trick guarantees the correlation we want!
asset_B_centered <- chosen_correlation * asset_A_centered +
  sqrt(1 - chosen_correlation^2) * independent_noise

# Step 4: Add a small average expected return to both (e.g., 0.05% daily)
asset_A_return <- asset_A_centered + 0.0005
asset_B_return <- asset_B_centered + 0.0005

# Check the generated correlation
cor(asset_A_return, asset_B_return)

plot(
  asset_A_return,
  asset_B_return,
  pch = 16,
  col = rgb(0, 0, 1, 0.35),
  main = "Simulated Asset Returns with Chosen Correlation",
```

```
xlab = "Asset A Daily Return",
ylab = "Asset B Daily Return"
)
abline(h = 0.0005, v = 0.0005, col = "gray")
```

Reflection Question: Correlation as a Story

In words, what does $\rho = -0.5$ mean? Does it mean that one asset always goes up when the other goes down? Why or why not?

Common Mistake

Correlation is not the same as causation. If two assets move together, it does not automatically mean one causes the other to move. There may be a shared economic factor, such as interest rates, inflation expectations or market sentiment.

2 The Diversification Engine: Portfolio Variance

Mini-Mission: Find the hidden engine of diversification

You will test how changing correlation changes portfolio risk, even when the individual assets stay the same.

Key Definition / Result

For a two-asset portfolio with weights w_A and $w_B = 1 - w_A$, expected return is

$$E[R_p] = w_A E[R_A] + w_B E[R_B].$$

Portfolio variance is

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho_{AB} \sigma_A \sigma_B.$$

The final term,

$$2w_A w_B \rho_{AB} \sigma_A \sigma_B,$$

is the **dependence term**. It is the mathematical reason why diversification may work or fail.

Suggested AI Prompt: Manual Portfolio Risk Function ** Intermediate

"Please help me write an R function that calculates the expected return and volatility of a two-asset portfolio. The function should take asset weights, expected returns, volatilities and correlation as inputs. Use clear variable names suitable for beginners."

Worked Example: Portfolio Mean and Volatility Function

**

Intermediate**R Code:**

```
rm(list = ls())

calculate_portfolio_stats <- function(weight_asset1, weight_asset2,
                                     return_asset1, return_asset2,
                                     volatility_asset1, volatility_asset2,
                                     correlation) {

  if (abs(weight_asset1 + weight_asset2 - 1) > 1e-6) {
    warning("Heads up: Your asset weights do not sum to 1 (100%).")
  }

  # Step 1: Expected Return
  expected_portfolio_return <- (weight_asset1 * return_asset1) + (weight_asset2 * return_asset2)

  # Step 2: Variance
  variance_asset1 <- volatility_asset1^2
  variance_asset2 <- volatility_asset2^2
  covariance <- volatility_asset1 * volatility_asset2 * correlation

  portfolio_variance <- (weight_asset1^2 * variance_asset1) +
    (weight_asset2^2 * variance_asset2) +
    (2 * weight_asset1 * weight_asset2 * covariance)

  # Step 3: Volatility
  portfolio_volatility <- sqrt(portfolio_variance)

  # Step 4: Output
  results <- list(
    Expected_Return = expected_portfolio_return,
    Volatility = portfolio_volatility
  )

  return(results)
}

my_portfolio <- calculate_portfolio_stats(
  weight_asset1 = 0.60,
  weight_asset2 = 0.40,
  return_asset1 = 0.10,
```

```

return_asset2 = 0.05,
volatility_asset1 = 0.15,
volatility_asset2 = 0.05,
correlation = 0.20
)

print(my_portfolio)

```

Predict Before Coding: Rank the four correlation worlds

Before calculating, rank +1, 0, -0.3 and -1 from highest portfolio risk to lowest portfolio risk.

Investigation / Activity: Changing Correlation While Keeping Everything Else Fixed

★★

Intermediate

Task. Compare portfolio volatility when correlation is +1, 0, -0.3, and -1.

R Code:

```

# 1. Define the different correlation values we want to test
correlation_values <- c(1, 0, -0.3, -1)

# 2. Create an empty vector to store the volatilities we calculate
portfolio_volatilities <- c()

# 3. Use a 'for' loop to run our calculation for EACH correlation value
for (cor in correlation_values) {

  # Call the function we created earlier
  temp_portfolio <- calculate_portfolio_stats(
    weight_asset1 = 0.60,
    weight_asset2 = 0.40,
    return_asset1 = 0.10,
    return_asset2 = 0.05,
    volatility_asset1 = 0.15,
    volatility_asset2 = 0.05,
    correlation = cor # <--- We plug in the current correlation from our loop here!
  )

  # Extract the Volatility from the result and add it to our storage vector
  portfolio_volatilities <- c(portfolio_volatilities, temp_portfolio$Volatility)
}

# 4. Combine the correlations and their resulting volatilities into a neat table
comparison_table <- data.frame(
  Correlation = correlation_values,

```

```
Volatility = portfolio_volatilities
)

# 5. Print the final comparison table
print(comparison_table)
```

Reflection Question: The Deep Idea

Why can correlation change portfolio risk without changing portfolio expected return? What does this tell us about the difference between return and risk?

3 Exploring the Efficient Frontier by R

Mini-Mission: Draw the map of risk and return

You will create a visual map of portfolio choices and compare how different correlations change the curve.

Suggested AI Prompt: Efficient Frontier Under Different Correlations

*** **Advanced**

"Act as an R programming mentor. I want to explore a two-asset efficient frontier. Asset A has expected return 12% and volatility 20%. Asset B has expected return 8% and volatility 10%. Test portfolio weights from 0% to 100% in 1% steps. Plot expected return against volatility. Repeat this for correlations +1, 0, -0.3 and -1. Use clear names and beginner-friendly comments."

Predict Before Coding: Which curve bends furthest left?

Before running the graph, predict which correlation should produce the strongest diversification benefit.

Investigation / Activity: Drawing Several Frontier Curves*** Advanced

R Code:

```

# If you don't have ggplot2 installed, you will need to run this line once:
# install.packages("ggplot2")

# Load the library so we can use its plotting functions
library(ggplot2)

# 1. Define our asset characteristics (using decimals)
return_asset_A <- 0.12
volatility_asset_A <- 0.20
return_asset_B <- 0.08
volatility_asset_B <- 0.10

# 2. Define the correlations and the sequence of weights to test
correlations_to_test <- c(1, 0, -0.3, -1)

# seq() creates a sequence of numbers. Here, from 0 to 1 in steps of 0.01 (1%)
weights_asset_A <- seq(from = 0, to = 1, by = 0.01)

# 3. Create an empty data frame (table) to store all our results
frontier_data <- data.frame(
  Volatility = numeric(),
  Expected_Return = numeric(),
  Correlation = character() # We treat correlation as a category/text for coloring the chart
)

# 4. Nested Loops: Calculate stats for EVERY weight, for EVERY correlation
for (cor in correlations_to_test) {

  for (weight_A in weights_asset_A) {

    # Since they must sum to 1, Asset B's weight is just 1 minus Asset A's weight
    weight_B <- 1 - weight_A

    # Calculate Expected Return
    port_return <- (weight_A * return_asset_A) + (weight_B * return_asset_B)

    # Calculate Volatility
    variance_A <- volatility_asset_A^2
    variance_B <- volatility_asset_B^2
    covariance <- volatility_asset_A * volatility_asset_B * cor

    port_variance <- (weight_A^2 * variance_A) +
      (weight_B^2 * variance_B) +
      (2 * weight_A * weight_B * covariance)

    port_volatility <- sqrt(port_variance)

    # Save this specific calculation as a temporary row
    temp_row <- data.frame(
      Volatility = port_volatility,
      Expected_Return = port_return,
      Correlation = as.character(cor) # Store as text so ggplot colors it cleanly
    )
  }
}

```

```

    )

    # Add this row to the bottom of our main table using rbind (Row Bind)
    frontier_data <- rbind(frontier_data, temp_row)
  }
}

# 5. Plot the Efficient Frontier using ggplot2
ggplot(data = frontier_data, aes(x = Volatility, y = Expected_Return, color = Correlation)) +
  geom_path(linewidth = 1) + # geom_path connects dots exactly in the order we calculated them
  labs(
    title = "Two-Asset Efficient Frontier",
    subtitle = "How correlation affects the risk/return tradeoff",
    x = "Portfolio Volatility (Risk)",
    y = "Expected Portfolio Return"
  ) +
  theme_minimal() # A clean, simple background theme for the chart

```

Common Mistake

A very beautiful efficient frontier does not automatically mean the portfolio is realistic. Real markets have transaction costs, liquidity constraints, estimation error, changing correlations and practical restrictions on short selling.

4 Minimum Variance Portfolio and Sharpe Ratio

Mini-Mission: Choose the best portfolio, but define “best” first

You will compare two ideas of optimality: lowest risk and best risk-adjusted return.

Key Definition / Result

The **minimum variance portfolio** is the portfolio with the lowest volatility among the tested weights.

The **Sharpe ratio** is

$$\text{Sharpe Ratio} = \frac{E[R_p] - R_f}{\sigma_p}.$$

It measures extra expected return per unit of volatility. A higher expected re-

turn is not automatically better if it requires too much risk.

Suggested AI Prompt: Minimum Variance and Maximum Sharpe Ratio

*** Advanced

"Please extend the two-asset frontier code. For a chosen correlation, calculate the portfolio volatility, expected return and Sharpe ratio for every weight. Identify the minimum variance portfolio and the maximum Sharpe ratio portfolio. Use a 3% risk-free rate."

Investigation / Activity: Comparing Two Kinds of Optimality

Advanced

R Code:

```
# 1. Define our asset characteristics and the risk-free rate
return_asset_A <- 0.12
volatility_asset_A <- 0.20
return_asset_B <- 0.08
volatility_asset_B <- 0.10
risk_free_rate <- 0.03 # 3% Risk-Free Rate

# Let's choose a correlation of 0 (no relationship) for this analysis
chosen_correlation <- 0

# 2. Create sequences for our weights (from 0 to 1 in steps of 0.01)
weights_asset_A <- seq(from = 0, to = 1, by = 0.01)

# Since weights must sum to 1, Asset B's weight is 1 minus Asset A's weight
# Notice we are subtracting an entire list of numbers from 1 all at once!
weights_asset_B <- 1 - weights_asset_A

# 3. Calculate metrics for ALL weight combinations simultaneously (Vectorization)

# Expected Return
portfolio_returns <- (weights_asset_A * return_asset_A) + (weights_asset_B * return_asset_B)

# Volatility
variance_A <- volatility_asset_A^2
variance_B <- volatility_asset_B^2
covariance <- volatility_asset_A * volatility_asset_B * chosen_correlation

portfolio_variances <- (weights_asset_A^2 * variance_A) +
  (weights_asset_B^2 * variance_B) +
  (2 * weights_asset_A * weights_asset_B * covariance)

portfolio_volatilities <- sqrt(portfolio_variances)
```

```

# 4. Calculate the Sharpe Ratio for every portfolio
# Formula: (Expected Return - Risk-Free Rate) / Volatility
sharpe_ratios <- (portfolio_returns - risk_free_rate) / portfolio_volatilities

# 5. Combine everything into a neat master table (Data Frame)
portfolio_data <- data.frame(
  Weight_Asset_A = weights_asset_A,
  Weight_Asset_B = weights_asset_B,
  Expected_Return = portfolio_returns,
  Volatility = portfolio_volatilities,
  Sharpe_Ratio = sharpe_ratios
)

# 6. Identify the Minimum Variance Portfolio (MVP)
# which.min() finds the exact row index where Volatility is the lowest
row_mvp <- which.min(portfolio_data$Volatility)
min_variance_portfolio <- portfolio_data[row_mvp, ] # Extract that specific row

# 7. Identify the Maximum Sharpe Ratio Portfolio (The Optimal Risky Portfolio)
# which.max() finds the exact row index where the Sharpe Ratio is the highest
row_max_sharpe <- which.max(portfolio_data$Sharpe_Ratio)
max_sharpe_portfolio <- portfolio_data[row_max_sharpe, ] # Extract that specific row

# 8. Print the final discoveries!
cat("\n--- Minimum Variance Portfolio (Lowest Risk) ---\n")
print(min_variance_portfolio)

cat("\n--- Maximum Sharpe Ratio Portfolio (Best Risk-Adjusted Return) ---\n")
print(max_sharpe_portfolio)

```

Worked Example: Visualising the Two Special Portfolios *** Advanced

R Code:

```

# Make sure the ggplot2 library is loaded
library(ggplot2)

# Create the visualization step-by-step
frontier_plot <- ggplot(data = portfolio_data, aes(x = Volatility, y = Expected_Return)) +

  # Layer 1: Draw the Efficient Frontier line
  geom_path(linewidth = 1, color = "darkblue") +

  # Layer 2: Add a large red dot for the Minimum Variance Portfolio (MVP)
  # Notice how we tell ggplot to temporarily use a different dataset for this dot!
  geom_point(data = min_variance_portfolio,
    aes(x = Volatility, y = Expected_Return),
    color = "red", size = 4) +

  # Layer 3: Add a large green dot for the Maximum Sharpe Ratio Portfolio
  geom_point(data = max_sharpe_portfolio,

```

```

aes(x = Volatility, y = Expected_Return),
color = "forestgreen", size = 4) +

# Layer 4: Add text labels next to our dots so we know what they are
# hjust controls horizontal alignment, and we shift the x/y slightly so the text doesn't overlap
the dot
annotate("text",
  x = min_variance_portfolio$Volatility + 0.003,
  y = min_variance_portfolio$Expected_Return,
  label = "Min Variance (Lowest Risk)", color = "red", hjust = 0, fontface = "bold") +

annotate("text",
  x = max_sharpe_portfolio$Volatility,
  y = max_sharpe_portfolio$Expected_Return + 0.002,
  label = "Max Sharpe (Best Trade-off)", color = "forestgreen", vjust = 0, fontface = "bold"
) +

# Layer 5: Add neat titles and axis labels
labs(
  title = "Efficient Frontier with Optimal Portfolios",
  subtitle = "Highlighting the Minimum Variance and Maximum Sharpe points (Correlation = 0)",
  x = "Portfolio Volatility (Risk)",
  y = "Expected Portfolio Return"
) +

# Layer 6: Apply a clean, professional theme
theme_minimal()

# Display the final plot!
print(frontier_plot)

```

5 AI-Assisted Interpretation: Good Prompting and Critical Checking

Suggested AI Prompt: Asking AI to Interpret Frontier Results

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Insight

"I have generated an efficient frontier for two assets in R. The minimum variance portfolio and maximum Sharpe ratio portfolio are different. Please explain in student-friendly language why these two portfolios can differ. Then suggest three reflection questions connecting the result to actuarial risk management."

Investigation / Activity: Critiquing An AI Explanation**** Insight**

Task. Ask an AI tool to explain the difference between minimum variance and maximum Sharpe ratio. Then check whether its answer includes all three of the following ideas:

1. Minimum variance focuses only on reducing volatility.
2. Sharpe ratio balances excess return against volatility.
3. The better portfolio depends on the decision-maker's objective and risk appetite.

Common Mistake

Do not ask AI only for “the answer”. Ask it to explain assumptions, define variables, comment the code, suggest checks, and identify possible weaknesses. In actuarial work, understanding the assumptions is often as important as producing the calculation.

6 Challenge Extension: When Diversification Fails

Mini-Mission: Stress test the comfort zone

You will check how the same portfolio can become riskier when correlation changes during stress.

Key Definition / Result

Diversification works best when risks are not perfectly positively correlated. It can fail when many assets become exposed to the same hidden factor. In market crises, assets that seemed different may suddenly move together.

Investigation / Activity: Stress Testing Correlation***** Challenge**

Task. Suppose a portfolio was designed under correlation -0.3 , but during a stress period the correlation becomes $+0.8$. Compare the volatility of the same portfolio weight under both assumptions.

R Code:

```
# 1. Define the volatilities (from our previous exercises)
volatility_asset_A <- 0.20
volatility_asset_B <- 0.10

# 2. Choose a portfolio weight to test (Let's use a 50/50 split)
weight_A <- 0.50
weight_B <- 0.50

# 3. Define our two scenarios
correlation_normal <- -0.3
correlation_stress <- 0.80

# 4. Calculate Variance and Volatility for the NORMAL scenario
covariance_normal <- volatility_asset_A * volatility_asset_B * correlation_normal
variance_normal <- (weight_A^2 * volatility_asset_A^2) +
  (weight_B^2 * volatility_asset_B^2) +
  (2 * weight_A * weight_B * covariance_normal)
volatility_normal <- sqrt(variance_normal)

# 5. Calculate Variance and Volatility for the STRESS scenario
covariance_stress <- volatility_asset_A * volatility_asset_B * correlation_stress
variance_stress <- (weight_A^2 * volatility_asset_A^2) +
  (weight_B^2 * volatility_asset_B^2) +
  (2 * weight_A * weight_B * covariance_stress)
volatility_stress <- sqrt(variance_stress)

# 6. Calculate the percentage increase in risk
risk_increase_pct <- (volatility_stress - volatility_normal) / volatility_normal * 100

# 7. Print the results clearly
cat("---- Portfolio Stress Test Results ----\n")
cat("Portfolio Weight: 50% Asset A, 50% Asset B\n\n")
cat("Normal Volatility (Correlation = -0.3):", round(volatility_normal * 100, 2), "%\n")
cat("Stress Volatility (Correlation = +0.8):", round(volatility_stress * 100, 2), "%\n\n")
cat("Risk Jump During Crisis: The portfolio volatility increased by", round(risk_increase_pct, 2),
    "%!\n")
```

Reflection Question: The Actuarial Mindset

Why is it dangerous to say, “This portfolio is safe because the historical correlation was low”? What could change in the future?

7 Real-World 4-Asset Portfolio & Monte Carlo Simulation

Mini-Mission: From classroom model to market experiment

This optional extension uses real data and random weights to explore many possible 4-asset portfolios.

In this session, we transition from theoretical 2-asset models to evaluating a 4-asset portfolio using real market data. We will calculate the portfolio's expected return and volatility without using complex matrix algebra, and then deploy a Monte Carlo simulation to uncover the Efficient Frontier.

7.1 Task 1: Evaluating a 4-Asset Portfolio Using Real Data

Using the `quantmod` package, we can pull historical stock data and calculate daily returns. By blending these individual returns into a single "Portfolio Daily Return" timeline, we completely bypass the need for a complex covariance matrix.

Listing 1: 4-Asset Portfolio Evaluation without Matrices

```
# Load the library to fetch financial data
library(quantmod)

# 1. Download real stock data (AAPL, MSFT, GOOG, AMZN)
getSymbols(c("AAPL", "MSFT", "GOOG", "AMZN"), from = "2023-01-01", to = "2024-01-01")

# 2. Extract Adjusted Close prices and calculate daily returns
# ROC stands for Rate of Change
returns_AAPL <- ROC(Ad(AAPL), type = "discrete", na.pad = FALSE)
returns_MSFT <- ROC(Ad(MSFT), type = "discrete", na.pad = FALSE)
returns_GOOG <- ROC(Ad(GOOG), type = "discrete", na.pad = FALSE)
returns_AMZN <- ROC(Ad(AMZN), type = "discrete", na.pad = FALSE)

# 3. Define equal weights (25% each, must sum to 1.0)
w_AAPL <- 0.25; w_MSFT <- 0.25; w_GOOG <- 0.25; w_AMZN <- 0.25

# 4. Calculate the DAILY portfolio return timeline
port_daily_returns <- (w_AAPL * returns_AAPL) + (w_MSFT * returns_MSFT) +
  (w_GOOG * returns_GOOG) + (w_AMZN * returns_AMZN)

# 5. Annualize Expected Return and Volatility (assuming 252 trading days)
annual_return <- mean(port_daily_returns) * 252
annual_vol <- sd(port_daily_returns) * sqrt(252)

# Print results
cat("Expected Annual Return:", round(annual_return * 100, 2), "%\n")
cat("Annualized Volatility (Risk):", round(annual_vol * 100, 2), "%\n")
```

7.2 Task 2: Monte Carlo Simulation for the Efficient Frontier

Calculating every exact weight combination for 4 assets is mathematically overwhelming. Instead, we use a Monte Carlo simulation to generate thousands of random weight combinations. The upper-left boundary of the resulting "cloud" of portfolios will naturally reveal the Efficient Frontier.

Listing 2: Monte Carlo Efficient Frontier Simulation

```
library(ggplot2)

# Define simulation parameters
num_portfolios <- 2500
sim_returns <- numeric(num_portfolios)
sim_volatilities <- numeric(num_portfolios)
sim_sharpes <- numeric(num_portfolios)
risk_free_rate <- 0.03

# Run the Monte Carlo Simulation
for (i in 1:num_portfolios) {

  # Generate 4 random numbers and normalize them so they sum to 1 (100%)
  rands <- runif(4, min = 0, max = 1)
  weights <- rands / sum(rands)

  # Calculate daily portfolio returns using the timeline approach
  daily_ret <- (weights[1] * returns_AAPL) + (weights[2] * returns_MSFT) +
    (weights[3] * returns_GOOG) + (weights[4] * returns_AMZN)

  # Annualize and store the risk/return metrics
  ann_ret <- mean(daily_ret, na.rm = TRUE) * 252
  ann_vol <- sd(daily_ret, na.rm = TRUE) * sqrt(252)

  sim_returns[i] <- ann_ret
  sim_volatilities[i] <- ann_vol
  sim_sharpes[i] <- (ann_ret - risk_free_rate) / ann_vol
}

# Combine simulated data into a Data Frame for Plotting
portfolio_cloud <- data.frame(Volatility = sim_volatilities,
                              Expected_Return = sim_returns,
                              Sharpe_Ratio = sim_sharpes)

# Plot the Efficient Frontier Cloud
ggplot(portfolio_cloud, aes(x = Volatility, y = Expected_Return, color = Sharpe_Ratio)) +
  geom_point(alpha = 0.5, size = 1.5) +
  scale_color_viridis_c(option = "plasma") +
  labs(title = "4-Asset Efficient Frontier (Monte Carlo Simulation)",
       subtitle = "Simulating 2,500 random tech portfolios",
       x = "Annualized Volatility (Risk)",
       y = "Annualized Expected Return",
       color = "Sharpe Ratio") +
  theme_minimal()
```

8 Consolidation Questions

Investigation / Activity: Review Set A: Risk Register and Scoring ★

Core

1. Write two new risk statements using the structure “If ..., then ...”.
2. Assign likelihood and impact scores from 1 to 3.
3. Explain why the risk score is useful but imperfect.
4. Identify one treatment for each risk.
5. Explain the difference between inherent risk and residual risk.

Investigation / Activity: Review Set B: Frequency and Severity ★★

Intermediate

1. Define frequency and severity in the claim simulation.
2. Explain why total loss is random.
3. Interpret the mean, standard deviation and 95th percentile of simulated total losses.
4. **Explain why a Uniform distribution might not be realistic for insurance severity (e.g., real claims have no strict upper limit and tend to be skewed).**
5. Describe one limitation of our Binomial frequency model.

Investigation / Activity: Review Set C: Correlation and Portfolio Risk ★★★

Advanced

1. State the two-asset portfolio variance formula.
2. Identify the covariance or correlation term in the formula.

3. Explain why lower correlation can reduce portfolio volatility.
4. Compare the meaning of minimum variance and maximum Sharpe ratio.
5. Explain how stress testing correlation helps risk management.

Investigation / Activity: Challenge Questions***** Challenge**

1. If two risks have correlation zero, are they necessarily independent? Explain.
2. Can diversification eliminate all risk in real financial markets? Why or why not?
3. Why might a low-return asset still appear in a good portfolio?
4. How could model error affect an efficient frontier?
5. How would transaction costs, liquidity risk or short-selling constraints change the decision?

End of Tutorial 6